A Boundary-Cap Counterexample to a Support-Point Question for Normal Loewner Chains

fa_banach_001 / agent_lane_00

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Status: candidate counterexample, likely valid. Question 3.10 is false as literally printed. The example is a bounded support point of S^0, but the Herglotz generator of a normal Loewner chain starting there is uniformly strictly dissipative near a boundary cap of positive surface measure.

1 Source question

Bracci and Roth ask the following on PDF page 6 of *Support points and the Bieberbach conjecture in higher dimension*, arXiv:1603.01532 [1].

Question 3.10. *Let $G(z,t)$ be a Herglotz vector field associated with the class $\mathcal{M}$ which generates a normal Loewner chain (f_t). Assume that*

$$\limsup_{z\to A} \mathsf{Re}\left\langle G(z,t), \frac{z}{\|z\|^2}\right\rangle \le -a,$$

for some $A \subset \partial\mathbb{B}^2$ and a.e. $t \ge 0$. Is it true that $f_0 \notin \mathsf{Supp}(S^0) \cup \mathsf{Ex}(S^0)$?

Figure 1: Question 3.10 rendered from page 6 of the source paper.

In transcription: if a Herglotz vector field $G(z,t)$ generates a normal Loewner chain (f_t) and

$$\limsup_{z\to A} \mathrm{Re}\left\langle G(z,t), \frac{z}{\|z\|^2}\right\rangle \le -a$$

for some $A \subset \partial\mathbb{B}^2$ and almost every $t \ge 0$, must $f_0 \notin \mathrm{Supp}(S^0) \cup \mathrm{Ex}(S^0)$?

2 Definitions

Let $\mathbb{B}^2$ be the Euclidean unit ball in $\mathbb{C}^2$. The source uses

$$\mathcal{M} = \{h \in \mathrm{Hol}(\mathbb{B}^2, \mathbb{C}^2) : h(0) = 0,\ dh_0 = \mathrm{id},\ \mathrm{Re}\langle h(z), z\rangle > 0\ (z \ne 0)\}.$$

A Herglotz vector field associated with $\mathcal{M}$ satisfies $-G(\cdot,t) \in \mathcal{M}$ almost everywhere. A normal Loewner chain has $f_t(0) = 0$, $d(f_t)_0 = e^t\,\mathrm{id}$, nested images, and a normal family $\{e^{-t} f_t\}$.

The compact class S^0 consists of normalized univalent maps admitting parametric representation. A map $f \in S^0$ is a support point if it maximizes the real part of a nonconstant continuous linear functional on S^0.

3 Proof intuition

The source question imposes strict dissipativity only near *some* boundary subset. A support-point generator can lose strict dissipativity at a small extremal locus while remaining uniformly strict on a large cap away from that locus. The sharp quadratic shear already present in the source does exactly this.

Its coefficient is chosen so that the generator is barely admissible: the dissipativity expression reaches zero at the points maximizing $|z_1||z_2|^2$. Near the coordinate circle $z_2 = 0$, however, the same expression is close to -1. A fixed cap around that circle therefore meets the hypothesis, even though the initial shear is detected as a support point by the sharp coefficient functional.

4 Counterexample

Set

$$c = \frac{3\sqrt{3}}{2}, \qquad \Phi(z_1, z_2) = (z_1 + cz_2^2, z_2), \qquad f_t = e^t\Phi,$$

and define the autonomous field

$$G(z_1, z_2) = (-z_1 + cz_2^2, -z_2). \tag{1}$$

Theorem 1. *The family $(f_t)_{t\geq 0}$ is a normal Loewner chain generated by G, and $f_0 = \Phi$ belongs to $\mathrm{Supp}(S^0)$. Nevertheless, with*

$$A = \{\zeta \in \partial\mathbb{B}^2 : |\zeta_2| \leq 1/4\}$$

one has

$$\limsup_{z\to A} \mathrm{Re}\left\langle G(z,t), \frac{z}{\|z\|^2}\right\rangle \leq -\frac{3}{4}$$

for every $t \geq 0$. Hence this is a counterexample to Question 3.10.

Proof. **Step 1: the field is admissible.** Write $r = \|z\|$, $x = |z_1|$, and $y = |z_2|$. Elementary one-variable optimization gives

$$\max_{x^2+y^2=r^2} xy^2 = \frac{2r^3}{3\sqrt{3}}. \tag{2}$$

Consequently, for $0 < r < 1$,

$$\mathrm{Re}\langle -G(z), z\rangle = r^2 - c\,\mathrm{Re}(z_2^2\overline{z_1}) \geq r^2 - cxy^2 \geq r^2 - r^3 > 0.$$

Also $-G(0) = 0$ and $d(-G)_0 = \mathrm{id}$. Thus $-G \in \mathcal{M}$.

Step 2: the normal chain. The Jacobian of Φ is

$$d\Phi_z = \begin{pmatrix} 1 & 2cz_2 \\ 0 & 1 \end{pmatrix},$$

and direct multiplication using (1) gives $d\Phi_z G(z) = -\Phi(z)$. Therefore

$$\frac{\partial f_t}{\partial t}(z) = f_t(z) = -d(f_t)_z G(z),$$

so (f_t) solves the Loewner–Kufarev PDE. Each f_t is an entire univalent triangular map, and $\{e^{-t}f_t\} = \{\Phi\}$ is normal. The standard Loewner existence/subordination theorem quoted in the source therefore makes (f_t) a normal Loewner chain. In particular, $\Phi = f_0 \in S^0$.

Step 3: Φ is a support point. We include the short coefficient argument used in the source. If $f \in S^0$ has coefficient $b^1_{0,2}$ and is represented by a Herglotz field with coefficient $q^1_{0,2}(t)$, the degree-two evolution equation has no lower-degree remainder and yields

$$b^1_{0,2}(f) = \int_0^\infty e^{-t} q^1_{0,2}(t)\, dt. \tag{3}$$

Harmonic decoupling preserves the field

$$(-z_1 + q^1_{0,2}(t) z_2^2, -z_2).$$

Its admissibility, together with (2) at $r = 1$, implies

$$|q^1_{0,2}(t)| \leq \frac{3\sqrt{3}}{2} = c$$

for almost every t. Equation (3) gives $|b^1_{0,2}(f)| \leq c$. The continuous coefficient functional is not constant on S^0, and $b^1_{0,2}(\Phi) = c$. Hence Φ maximizes its real part and belongs to $\mathrm{Supp}(S^0)$.

Step 4: strict dissipation on a boundary cap. For $z \neq 0$,

$$\mathrm{Re} \left\langle G(z), \frac{z}{\|z\|^2} \right\rangle = -1 + c \frac{\mathrm{Re}(z_2^2 \overline{z_1})}{\|z\|^2}. \tag{4}$$

The set A is a nonempty closed cap with nonempty relative interior and positive surface measure. Passing to A in (4) gives

$$\limsup_{z \to A} \mathrm{Re} \left\langle G(z), \frac{z}{\|z\|^2} \right\rangle \leq -1 + c \sup_{\zeta \in A} |\zeta_1||\zeta_2|^2 \leq -1 + \frac{c}{16} = -1 + \frac{3\sqrt{3}}{32} < -\frac{3}{4}.$$

Because G is autonomous, this holds at every time. The hypothesis of Question 3.10 is satisfied with $a = 3/4$, but its conclusion is false since $f_0 = \Phi \in \mathrm{Supp}(S^0)$. □

5 Verification

The script `code/verify_counterexample.py` samples the exact one-variable maximization, checks the Loewner PDE identity at complex points, integrates the degree-two coefficient kernel, and generates the figure. It reports the exact extremal product to numerical precision, a zero PDE residual, a cap supremum below $-3/4$, and `PASS`. Full commands and output are recorded in `verification.md`; computation is not used as proof.

6 Scope and bounded novelty check

The theorem disproves Question 3.10 as printed, specifically its phrase "for some $A \subset \partial\mathbb{B}^2$." It does not address a strengthened question requiring $A = \partial\mathbb{B}^2$, or an unprinted largeness or global accessibility condition.

The run indexes were searched by arXiv id, title, question number, and core terms. A bounded primary-source web/arXiv search used the exact question language, author names, "exponentially squeezing," and the constant $3\sqrt{3}/2$. It found the source and nearby Loewner work but no later explicit resolution or matching cap counterexample. This is a bounded search, not an exhaustive novelty claim.

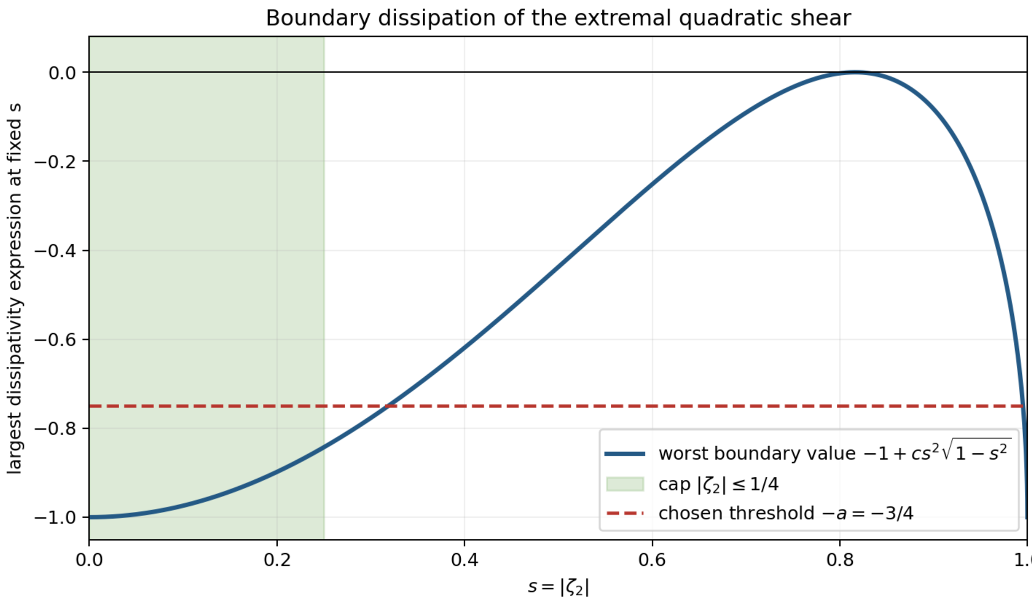

Figure 2: The worst boundary value at fixed $|\zeta_2|$. The green cap lies strictly below the chosen threshold $-3/4$.

7 Human-review recommendation

Independent review should focus on the intended semantics of $\limsup_{z\to A}$ and whether the authors silently intended a stronger hypothesis on A. Under the literal printed statement, the cap used here is substantial: it has positive surface measure and nonempty relative interior. The algebraic and coefficient checks are otherwise elementary and explicit.

References

[1] F. Bracci and O. Roth, *Support points and the Bieberbach conjecture in higher dimension*, Complex Analysis and Dynamical Systems, Trends in Mathematics, Birkhäuser/Springer (2018), 67–79; arXiv:1603.01532.